

2025 Ch H2 Q6

Section: Chemical Reactions

Topic: Rates of Reaction

Question Summary:

This question covers ion electron equations, combining redox half equations, dilution tables, rate graphs, and collision theory.

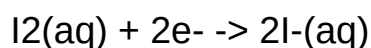
Worked Solution

(a) Reduction ion electron equation

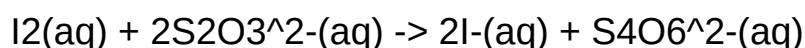


(b) Overall redox equation

Half equations:



Combined:



(c)(i) Volume of water in experiment 3

$$25 - 15 = 10 \text{ cm}^3$$

(c)(ii) Why must the beaker be dry?

Water in the beaker would dilute the reactants and change the rate.

(d)(i) Effect of doubling concentration

Doubling concentration roughly doubles the rate. First order in iodide.

(d)(ii) Time at concentration 0.025 mol l^{-1}

Rate approx 0.05 s^{-1} , so time = $1/0.05 = 20 \text{ s}$.

(d)(iii) Why increasing concentration increases rate

Higher concentration gives more particles per unit volume, so more collisions per second. More frequent successful collisions increase the rate.

Revision Tips

- When combining half equations, electrons must cancel.
- Rate = $1/t$ only works with the same visible endpoint.
- Dry glassware prevents accidental dilution.
- First order means rate is directly proportional to concentration.

- Collision frequency increases with concentration.